



EARTHQUAKE PREDICTION THROUGH QUANTUM MACHINE LEARNING

Brought to you by 0xDE4DB33F

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About us

We are a computer science major, two computer engineers, and an electrical engineer, who wear tasked with applying Quantum computer and machine learning with the goal of predicting earthquakes.

INTRODUCTION

DATA WORTH- WHILE



- Distance from fault/activity
- Distance from coastline
- Depth of occurrence
- ETAS

CHARLESONIAN ETAS

01.

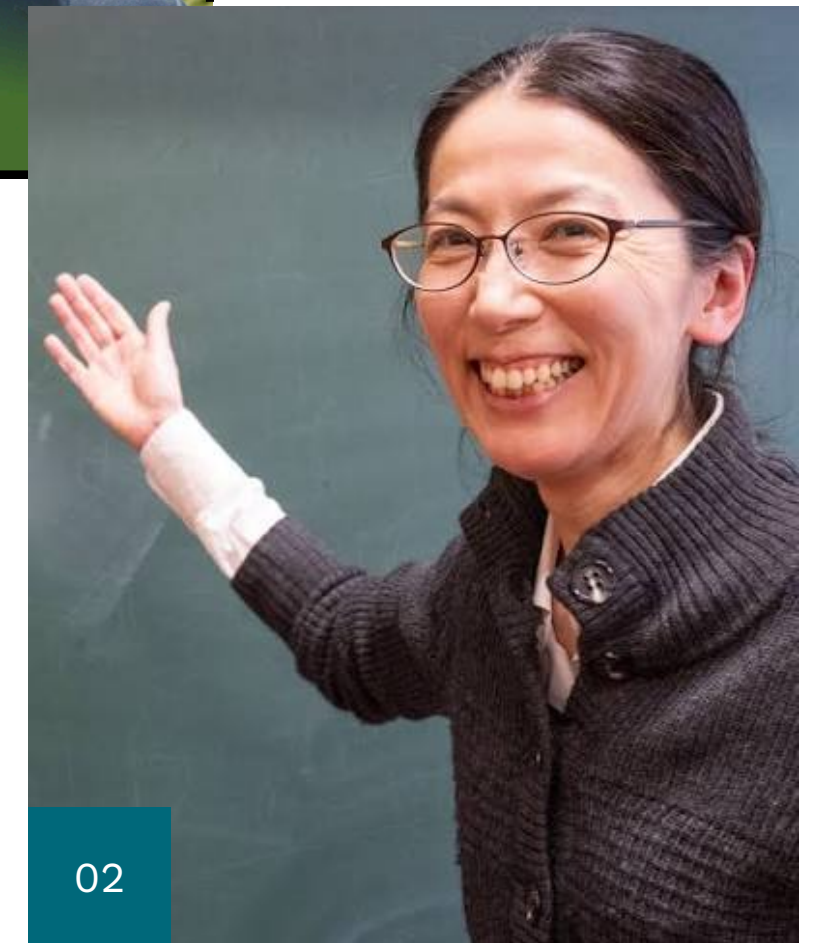
ETAS Implementation

Computes probability of aftershocks as a function of background quake rate and a prior rate from past data decaying with time.

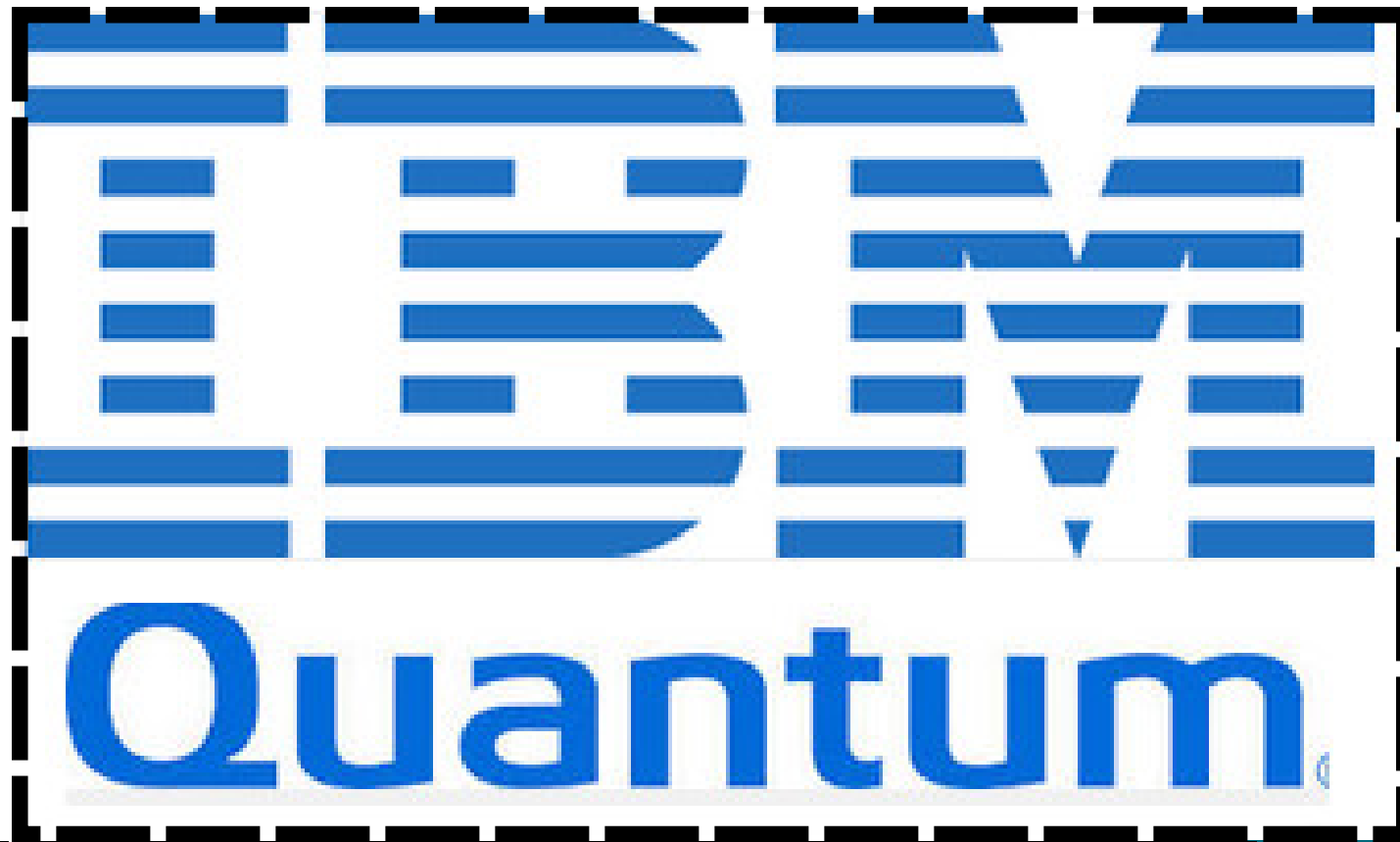
02.

Charlesonian

Outputs 5 parameters, each parameter is the sum of magnitudes occurring within a given distance from a point across 5 years of historical data



Yoshiko Ogata, creator of ETAS



ADDING QUANTUM

01

Classical Version

Slightly more effective than base-line model provided, suffered from correlated data points with poor stability

02

Pure Quantum

Lacked accessibility, ran much faster when quantum devices were available, handled more data

03

Hybrid Version

Used both a quantum simulator and real quantum device for hardware verification, best of both options

Final Results



- AUC: 0.8749
- ECE: 0.0499
- Accuracy: 0.7865
- Precision: 0.8182
- Recall: 0.8431
- F1 score: 0.8305
- Confusion matrix:

[[195 86]

[72 387]]

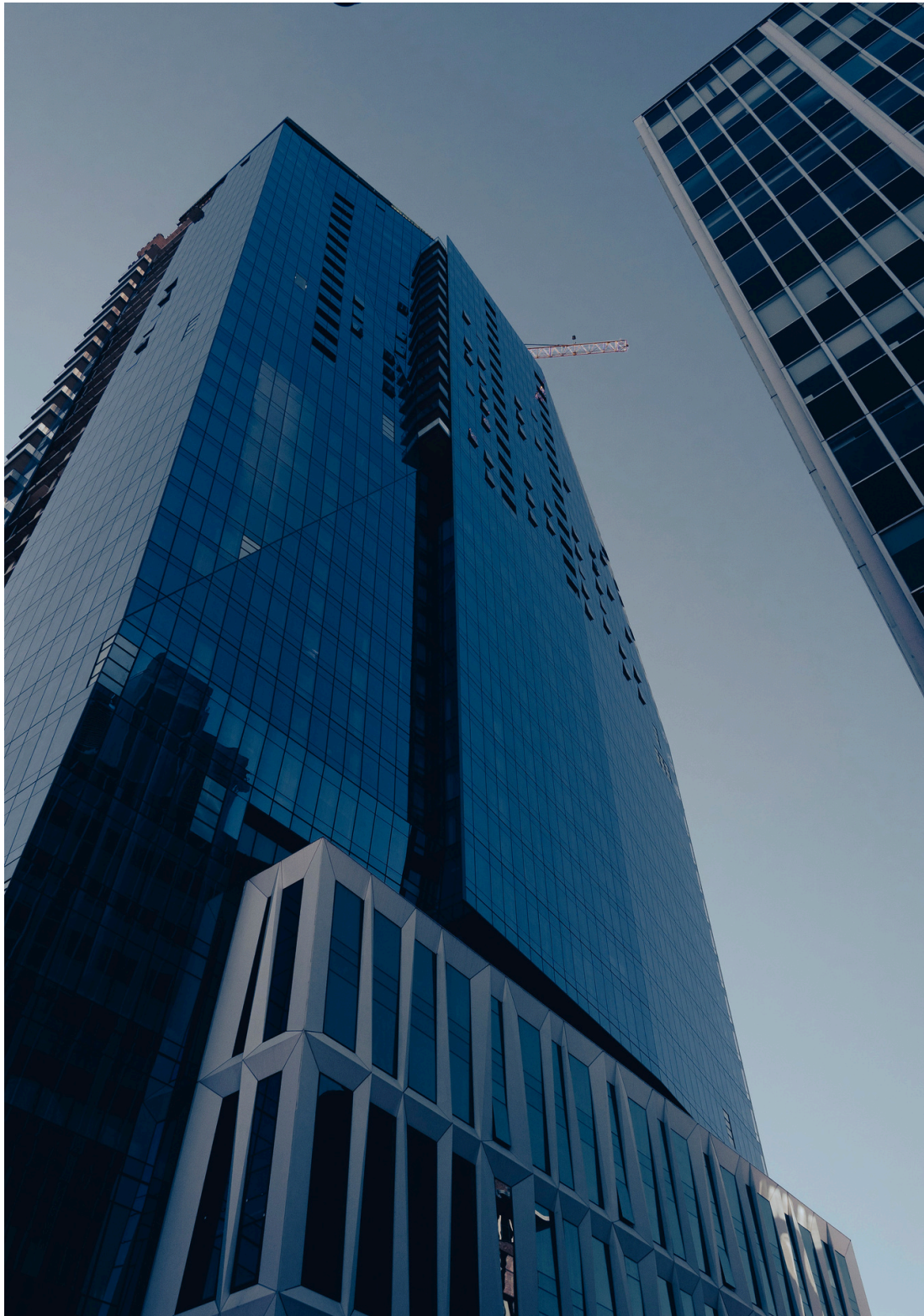
CLOSING REMARKS



Department of **ENGINEERING AND SCIENCE EDUCATION**

Overall, we saw an increase in our information bits and an overall decrease in our ECE. We all gained valuable experience in ML and Quantum during this experience.





SOURCES

- Chu, A., et al. “Comparison of Etas parameter estimates across different global tectonic zones.” Bulletin of the Seismological Society of America, vol. 101, no. 5, 26 Sept. 2011, pp. 2323–2339, <https://doi.org/10.1785/0120100115>.
- Public USGS Fault Data
- Provided Slide Decks
- Provided Data Sets



THANK YOU

